

MuTau channel: Combine tool

13/04/2026

Set a 95% CL upper limit on the signal strength $r = \sigma_{\text{obs}}/\sigma_{\text{SM}}$

data pipeline (BDT-based)

```
save_shapes.cpp    TMVAClassification.C    TMVAClassificationApplication.C  
(kinematic hists) --> (BDT training)      --> (BDT score per event)
```

↓
MuTau_shapes.root

↓
mutau_bdt_shape.txt --> mutau_bdt_workspace.root

↓
combine (Limits, Signif
FitDiagnostics, GoF, Im

corrections applied: **per-arm proton selection** and **signal cross-section normalization** (both bugs fixed before final results).

Bug fix 1: signal proton selection

Problem

The signal code required exactly two total proton entries, discarding events with ≥ 3 pileup protons.

Fix

Replaced with a per-arm requirement: **at least one valid proton in arm 0 AND at least one in arm 1.**

results

	Before	After	Change
Selected signal events (unweighted)	14,085	21,398	+7,313
Final weighted signal yield	0.0633	0.0794	+25%

bug 2: signal cross-section normalization

problem

The signal was normalized as:

$$w_{\text{sample}} = \frac{54900 \times w_{\text{SM}}[0]}{4000 \times 1000}$$

The denominator 4,000,000 has no physical basis in the sample.

Opened the upstream ntuple

(GammaGammaTauTau_2018_UL_MuTau_SMandBSMweights_ntuplesfromminiJuly.root): - Total events: **299,979** (no upstream filter) - $\sum w_{\text{SM}}[0]$ over all events: **402.661**

fix

Correct formula using $\sigma_{\text{SM}}^{\text{rwgt51}} = 0.0047$ pb:

$$w_{\text{sample}} = \frac{54900 \times 0.0047 \times w_{\text{SM}}[0]}{402.661}$$

bug 2: signal cross-section normalization (cont)

problem

The signal was normalized as:

$$w_{\text{sample}} = \frac{54900 \times w_{\text{SM}}[0]}{4000 \times 1000}$$

The denominator 4,000,000 has no physical basis in the sample.

verification

$$\sigma_{\text{implied}} = \frac{\sum w_{\text{sample}}}{L} = \frac{258.03}{54900} = 0.0047 \text{ pb} = 4.7 \text{ fb} \quad \checkmark$$

1: save_shapes.cpp

Produces MuTau_shapes.root – input ROOT file with per-sample kinematic histograms that Combine reads as shapes.

Input samples

Sample	File
Data	Data_2018_UL_MuTau_nano_merged_proton_vars.root
DY	DY_2018_UL_MuTau_nano_merged_pileup_protons.root
ttbar	ttjets_2018_UL_MuTau_nano_merged_pileup_protons.root
QCD	QCD_2018_UL_MuTau_nano_merged_proton_vars_new.root
Signal	MuTau_sinal_SM_2018_july.root

What it does

- 1 Requires `sist_mass > 0`; for MC: protons on both arms (`xi1 > 0 && xi2 > 0`)
- 2 Fills invariant mass (`m_X`, 30 bins, 0–1200 GeV) and 7 other kinematic histograms per sample
- 3 Applies MC normalization (DY: `Scale(1.004e-4)`, ttbar: `Scale(1.0)`, data/QCD: `weight = 1`)

2: TMVA (TMVAClassification.C)

Variable list

Central-system rapidity, acoplanarity, total system p_T , muon p_T , tau p_T , invariant mass, PPS mass-matching term, MET, rapidity-matching term.

BDT scan results (corrected normalization, $\sigma_{\text{SM}} = 4.7 \text{ fb}$)

Scan	Config	Exp. -2σ	Exp. -1σ	Exp. median	Exp. $+1\sigma$	Exp. $+2\sigma$	Observed
A	AdaBoost, $N = 600$, $d = 2$	0.79	1.20	2.04	3.76	6.18	6.03
B	AdaBoost, $N = 850$, $d = 3$	0.80	1.23	2.13	3.95	6.44	4.73
C	GradBoost, $N = 600$,	48.8	69.9	111.5	194.2	337.9	219.4

Methods trained

Likelihood, Fisher, BDT – **BDT selected as best discriminator.**

3: datacard (mutau_bdt_shape.txt)

channel and process structure

```
imax 1    # 1 channel: mutau
jmax 3    # 3 backgrounds: DY, ttbar, QCD
kmax *    # all nuisance parameters
```

shape mapping (all from ../MuTau_shapes.root)

Process	Histogram	Index
data_obs	bdt_data	–
sinal	bdt_sinal	0 (signal)
dy	bdt_dy	1
ttjets	bdt_ttjets	2
qcd	bdt_qcd	3

all rates set to -1 → Combine reads normalization directly from histogram integrals.
yield: **1,708 events** (protons on both arms).

3: systematic uncertainties

Lognormal (lnN) uncertainties

source	signal	dY	ttbar	QCD
Luminosity	1.5%	1.5%	1.5%	–
tau ID vs jet	5.0%	4.0%	4.0%	–
tau ID vs e	2.5%	2.5%	2.5%	–
tau ID vs mu	3.1%	3.2%	2.6%	–
Muon trigger	0.5%	0.07%	0.07%	–
Muon ID+Iso	0.02%	0.04%	0.04%	–
Muon reco	0.06%	0.05%	0.05%	–
Proton ξ (ξ_{Proton})	50%	50%	50%	50%

3: systematic uncertainties (cont)

Floating normalization factors (rateParam, range [0, 5])

- `beta_dy` for DY, constrained by `dy_norm` lnN 1.30
- `beta_tt` for `ttbar`, constrained by `ttjets_norm` lnN 1.50
- `beta_qcd` for QCD, constrained by `qcd_norm` lnN 1.50

4: running Combine

##Convert datacard to RooWorkspace

```
text2workspace.py mutau_bdt_shape.txt -o mutau_bdt_workspace.root -m 120
```

##Asymptotic 95% CL limits

```
combine -M AsymptoticLimits mutau_bdt_workspace.root -m 120 -n MuTau_bdt --cl 0.95
```

##Observed and expected significance

```
combine -M Significance mutau_bdt_workspace.root -m 120 -n MuTau_bdt --significance
```

```
combine -M Significance mutau_bdt_workspace.root -m 120 -n MuTau_bdt_exp -t -1 --expected
```

##Fit diagnostics

```
combine -M FitDiagnostics mutau_bdt_workspace.root -m 120 -n MuTau_bdt \
    --saveShapes --saveWithUncertainties --rMin 0 --rMax 500 --robustFit 1
```

##Best-fit signal strength(singles)

```
combine -M MultiDimFit mutau_bdt_workspace.root -m 120 -n MuTau_bdt_singles \
    --algo singles --rMin 0 --rMax 500 --robustFit 1
```

5: impacts

```
combineTool.py -M Impacts -d mutau_bdt_workspace.root -m 120 \  
  --doInitialFit --robustFit 1 --rMin 0 --rMax 500
```

```
combineTool.py -M Impacts -d mutau_bdt_workspace.root -m 120 \  
  --doFits --robustFit 1 --rMin 0 --rMax 500
```

```
combineTool.py -M Impacts -d mutau_bdt_workspace.root -m 120 -o impacts_MuTau_bdt.json
```

```
plotImpacts.py -i impacts_MuTau_bdt.json -o impacts_MuTau_bdt
```

results: 95% CL upper limits on r

Method: AsymptoticLimits, BDT shape, $m_H = 120$ GeV (Scan A, corrected normalization)

Estimate	$r = \sigma/\sigma_{\text{SM}}$	Absolute σ_{95}
Expected -2σ	—	—
Expected -1σ	—	—
Expected (median)	2.04	\$ \$9.6 fb
Expected $+1\sigma$	—	—
Expected $+2\sigma$	—	—
Observed	6.03	\$ \$28.3 fb

mass-shape cross-check: expected $r < 22.75$, observed $r < 45.0$ (mass as discriminant).
The BDT result is \$ 11\times\$ better than the mass-shape result

significance and best-fit signal strength

after normalization fix ($\sigma_{\text{SM}} = 4.7 \text{ fb}$)

Quantity	Result
Observed local significance	2.41σ
Expected significance at $r = 1$	1.62σ
Best-fit signal strength	$r = 2.13^{+1.92}_{-1.17}$

before fix (wrong normalization): expected significance was $\sim 0.046\sigma$, best-fit $r \sim 100$ – those numbers were artifacts of the $47\times$ normalization underestimate.

now: The mild excess (2.41σ) is consistent with a statistical fluctuation

goodness of fit and nuisance impacts

saturated GoF (after fix)

Quantity	Value
Observed test statistic	13.86
p-value	0.51

Observed value is well within the toy distribution-> the fitted model is compatible with data.

dominant nuisance impacts on r

Nuisance / group	Approx. effect on r
xiProton	+49 / - 33
beta_qcd	-24 / + 37
beta_dy	-22 / + 31
beta_tt	-21 / + 27
Muon / tau efficiency	small to moderate

Summary and next steps

corrected results (Scan A BDT, $\sigma_{SM} = 4.7$ fb)

Quantity	Value
Expected limit (median)	$r < 2.04$
Observed limit	$r < 6.03$
Observed significance	2.41σ
Expected significance	1.62σ
Best-fit r	$2.13^{+1.92}_{-1.17}$
GoF p-value	0.51

Scan B gives expected $r < 2.13$ (+4% vs A) – Scan A robustness confirmed.

Summary and next steps (cont)

What was achieved

- **bug 1 fixed:** proton selection corrected to per-arm logic (\$ \$1 per PPS arm): +25% signal yield
- **bug 2 fixed:** signal normalization corrected to $\sigma_{\text{SM}} = 4.7 \text{ fb}$ (was $47\times$ too small)
- BDT trained and scanned (A, B, C); Scan A selected ; Scan C rejected (unstable)
- statistical validation: limits, significance, fit diagnostics, GoF, impacts

Possible improvement items

- 1 **Post-fit shape validation:** identify which BDT bins drive the 2.41σ excess
- 2 **Proton modeling:** `xiProton` is the dominant nuisance – refine proton ξ uncertainty ?
- 3 **Background normalization:** assess DY, QCD, $t\bar{t}$ with control regions ?